FILIPE RAMOS

Computer Science Graduate (2027)

Southport, UK · 07593258165 · filiperamos212@gmail.com · github.com/Dialexy

PROFILE

Final-year Computer Science undergraduate at Edge Hill University who builds and rigorously evaluates machine learning systems end to end, from a hybrid retrieval-augmented generation pipeline achieving 97% recall to an in-progress NL2SQL research project. Strong on systems design and evaluation methodology, with experience communicating technical work clearly through six months of software teaching.

PROJECTS

RAG Pipeline

github.com/Dialexy/RAG-pipeline

- Built a hybrid retrieval-augmented generation system from scratch over a 10,000-article corpus: BM25 + dense fusion, LLM query expansion, cross-encoder reranking, confidence-gated abstention
- Achieved Recall@10 0.97, MRR 0.909, 100% Hit@3, 0.938 faithfulness on a held-out 100-question set; 99% abstention accuracy on out-of-corpus questions
- Diagnosed and fixed a chunk-indexing bug that had silently left 31% stale duplicates in the vector index, lifting MRR from 0.784 to 0.909 after the rebuild
- Ran a full component ablation study and built a 26-test automated suite validating each pipeline stage

NL2SQL Pipeline

Final Year Project (in progress)

- Designing a schema-linking natural-language-to-SQL system over a normalised 8-table PostgreSQL schema of arXiv papers, with a self-repair loop for query correction
- Building on Spider, RAT-SQL and DIN-SQL approaches to schema linking

EXPERIENCE

Assistant Coach, Jam Coding

March 2026 - Present · Ormskirk

- Co-facilitate after-school block-coding workshops for primary-age pupils, including direct teaching of core concepts (variables, loops, conditional logic) when needed
- Spend most of each session on 1:1 troubleshooting; on a typical hour, answering around 40 - 50 individual questions as pupils work through problems
- Adapt explanations for a young, non-technical audience, building the ability to break down ideas clearly and patiently

EDUCATION

BSc (Hons) Computer Science, Edge Hill University

2024 - 2027 (expected)

- Final-year project: NL2SQL pipeline over an academic paper database, with schema linking and a self-repair loop (see Projects)
- Relevant coursework: advanced databases, statistical modelling, secure complex systems

A-Levels, King George V College

2022 - 2024

- Chemistry, Biology, Mathematics

SKILLS

Languages: Python, C++, Java, SQL, TypeScript, Lua

Machine Learning / LLM Engineering: Hybrid retrieval (BM25 + dense fusion), cross-encoder reranking, embedding models (sentence-transformers), local LLM serving (Ollama), evaluation design (recall, MRR, faithfulness metrics, ablation studies), ChromaDB

Backend & Tools: FastAPI, PostgreSQL, Git, Docker, Linux, CMake, pytest

Web: React, TailwindCSS

Concepts: OOP, data structures & algorithms, dynamic programming, testing & debugging, version control, agile methodologies

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LANGUAGES & INTERESTS

English - Native **Portuguese** - Native

Interests: Gym, hiking, indoor rock climbing, gaming